

# Designing for Demoulding

Parametric Moulds that Know When to Let Go

Louis Sabri,

CODE3201 Graduation Project Theory

[louis.sabri@gmail.com](mailto:louis.sabri@gmail.com)  
[louissabri.github.io](https://louissabri.github.io)



Figure 1 - PETG mould with breakable seam

## Abstract.

Complex free-form geometries are increasingly viable in architecture, but traditional mould fabrication remains unsustainable and labour-intensive. Large-Format Additive Manufacturing (LFAM) presents a promising alternative for casting architectural elements. However, demoulding these intricate forms is a persistent challenge. This research adopts a design-through-research methodology that blends iterative prototyping, material experimentation, and computational design to address this issue. A parametric workflow integrated with LFAM technology using an MD10 pellet extruder mounted on a KUKA KR70 guides the development of moulds designed for controlled demouldability. Techniques explored include under-extrusion via dynamic speed control to create fracture points, toolpath deformation to embed tearable seams, and weaving extruder on/off commands to form perforations. The physical experiments conducted inform the refinement of digital workflows, identifying effective print parameters and evaluating structural stability. The outcome is a set of validated design strategies and fabrication practices optimised for demoulding performance without compromising form or function. The project contributes scalable, sustainable solutions to LFAM mould production, offering practical workflows for designers and fabricators while paving the way for further innovation in material systems, automated demoulding, and industrial applications.

### **Keywords:**

Cast Concrete, Demoulding, Grasshopper, LFAM, Robotic Fabrication

## Introduction.

Since the early 2000s advancements in 3D modelling software, increased computing power, and lower hardware costs have made complex free-form geometries a more viable option in design and architecture (Shelden, 2002; Pottmann, 2010; Pottmann et al., 2015). Traditional mould fabrication techniques to realise these complex forms however remain labour intensive, costly, and environmentally unsustainable (Li et al., 2022). Post curing, removing moulds with complex geometries can damage the cast component and prevent mould reuse. LFAM produced moulds offer a promising alternative for casting bespoke facade panels, columns, and other architectural elements, improving sustainability, cost efficiency, and fabrication accuracy (Armstrong et al., 2023; Hunt, 2023). However, demoulding these complex forms remains a critical, under explored challenge.

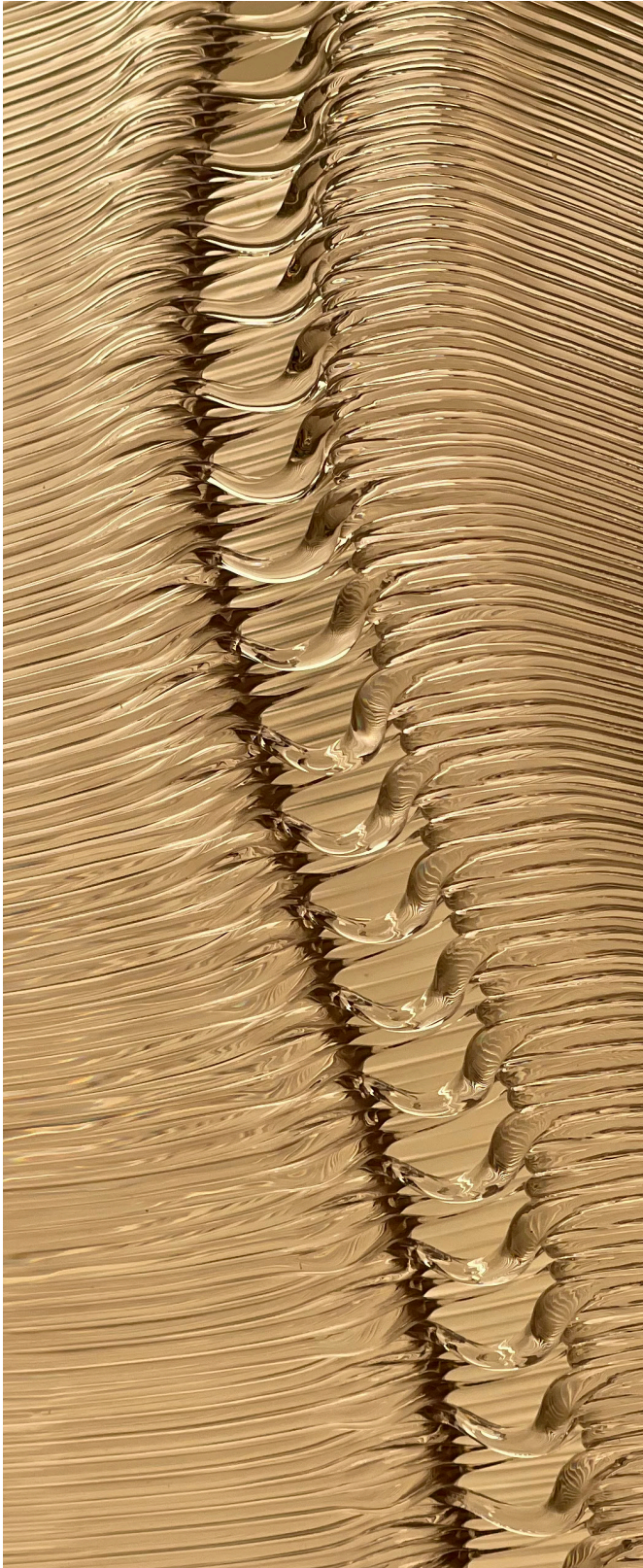
Design research, as an overarching methodology, frames this project by positioning iterative making as a primary mode of inquiry. Rather than separating theory from practice, research-through-design operates through cycles of prototyping, testing, and reflection, allowing insights to emerge directly from the act of designing. This approach is particularly suited to exploring the demoulding problem, as it demands empirical engagement with both material behaviour and digital workflows. By embedding experimentation within the design process, the project generates situated knowledge that bridges CAD, fabrication constraints, and practical demoulding strategies in a single integrated framework.

Within this methodology, a parametric design-

to-fabrication workflow integrated with LFAM technology guides iterative prototyping and analysis. Specifically, the research project adopts an MD pellet extruder mounted on a KUKA KR70 robotic arm. We investigate dynamic control of extrusion via movement speed to intentionally under extrude material and create predetermined fracture points, and structured weaving of extruder on/off commands to further influence extrusion & material deposition rates. Empirical data from iterative prototyping is collected to identify optimal printing parameters and evaluate structural stability and demoulding efficiency. The research contributes a refined parametric workflow and validates best practice optimisation for demouldability while maintaining structural integrity and reducing material deformation. The research delivers practical, scalable solutions for LFAM produced moulds, providing more sustainable and economically viable fabrication techniques for complex architectural forms. By developing a replicable parametric workflow for designers and fabricators, the project aims to accelerate the adoption of sustainable additive manufacturing practices across the architecture, engineering and construction (AEC) sector.



Figure 2 - PETG mould with breakable seam



## Research Aims.

To develop and validate a parametric design-to-fabrication workflow for LFAM-produced moulds that integrates demoulding strategies directly into the design process.

To experimentally test and refine demoulding techniques—such as dynamic under-extrusion, toolpath deformation, and extrusion perforations—for optimising mould removability without compromising form or function.

To contribute scalable, sustainable, and practically applicable fabrication strategies for complex concrete moulds, accelerating the adoption of LFAM in architectural construction.

## Research Question.

How can a demoulding strategy be integrated into a parametric design-to-fabrication workflow to 3D print mould geometries for in-situ & prefab cast concrete construction?



## Background Research.

The rise of complex geometries in contemporary architecture, often referred to as free-form structures, is primarily attributable to advancements in digital technology and computational design (Shelden, 2002; Pottmann, 2010; Pottmann et al., 2015). The evolution of free-form geometry has redefined traditional architectural practice, shifting the emphasis towards a more fluid and adaptable design mentality. Coupled with technological advancements and interdisciplinary collaboration, this mentality influences aesthetic preferences, and addresses broader social and environmental concerns within the architectural landscape. The integration of free-form geometries allows for optimised structural performance, and improved energy efficiency in buildings & construction processes, particularly in high-rise structures that are known for their energy demands (Shawkat et al., 2023; Yahya, 2023).

Moulds are a traditional and very commonly used method for fabricating free-form structures. Traditional formwork systems are usually fabricated out of wood or metal, manually constructed by skilled craftspeople. The fabrication process for traditional formworks is well established, but time consuming and expensive. Moulds for complex free-form geometries require a great deal of time, skill, and experience to fabricate, and typically produce a lot of waste material (Li et al., 2022). Modern advanced manufacturing methods have enabled faster and more precise processes for fabricating free-form mould geometries. Computer numerical control (CNC) milling of foam moulds is a current technique for

fabrication of unique free-form moulds, allowing for a much faster milling process and lower material costs (Chen, Shan and Dong, 2013; Xu et al., 2022). However, subtractive manufacturing techniques such as this are not without their drawbacks. CNC milling of any material produces a large amount of waste product, which is often difficult to recycle. If the free-form structure requires unique mould geometries, the milled material is limited to single use, and the materials employed are generally non-recyclable.

LFAM of free-form mould structures is a promising emerging technique that allows for the fabrication of complex free-form moulds more efficiently and with less waste (Hunt, 2023). A recent study by Armstrong et al. performed a lifecycle and economic analysis, finding that 3D printed fibre reinforced polymer moulds can be fabricated faster than traditional formwork, and reused many times – making them more cost effective. The formwork material can also be reground and recycled, reducing waste and carbon emissions.

LFAM of free-form mould structures is not without its own limitations, there are a wide range of technical challenges that must be addressed. Large format polymer extrusion prints tend to warp or delaminate if not controlled. Burger et al. found that a 2.5m long formwork printed in polypropylene (PP) warped by 6cm, and Polyethylene Terephthalate Glycol (PETG) by 2cm before failing, whereas a Polylactic Acid (PLA) print warped only ~1.3cm (Burger et al., 2022). Even so, very large and thin shells can suffer

layer delamination, which was noted as “a major challenge” during a 26 hour print of a full scale column formwork (Burger et al., 2020). Structural stability under concrete pressure is another concern of large-scale 3D printed formworks, as they are often thin to save material and thus struggle to resist the hydrostatic pressure of fresh concrete. Without reinforcement, a printed formwork can bulge or buckle significantly. One 40cm tall PLA formwork expanded up to 36% in width at the top during casting. Adding integrated stiffening ribs and securely anchoring the formwork can drastically reduce deformation down to ~4% expansion (Burger et al., 2022).

Chemical interactions between the formwork material and the alkaline environment of wet concrete can weaken certain 3D printed plastics, as the high pH causes cracking in susceptible polymers like PLA, leading to rapid strength loss. Jipa et al. found that PLA can lose over 80% of its strength after just 10 minutes in contact with alkaline solution, becoming prone to brittle failure. Other thermoplastics (such as Acrylonitrile Butadiene Styrene (ABS) or PETG) fare better, with little or no degradation under the same conditions. These findings guide material selection. For example, while PLA prints with less warp, its poor alkaline resistance is a serious concern when formworks will come into contact with fresh concrete (Jipa et al., 2022).

Once the concrete cures, stripping the plastic formwork can be laborious and may risk damaging intricate details. In most cases the printed moulds are sacrificed after use, raising sustainability concerns. However, Burger

et al. demonstrated that a PETG formwork could be shredded and re-extruded into new formwork. A “circular formwork” workflow was validated where the used plastic mould was removed, ground down, and reprinted into another mould – with little loss in material performance over at least one recycle cycle (Burger et al., 2023). Xu et al. proposed an innovative approach, printing moulds out of Thermoplastic polyurethane (TPU) to aid in the demoulding process. TPU is a much more flexible material than other 3D printed plastics. This flexibility was found to significantly alleviate the difficulties of the demoulding process. In addition, the TPU moulds remained intact after use – meaning it did not need to be ground down and reprinted for reuse (Xu et al., 2022). However, this approach may not be applicable for more complex, free-form geometries.

The aesthetic implications and surface finish of casting concrete in LFAM-produced moulds is another point of consideration. Naboni and Breseghello (2020) emphasise that high-resolution additive formwork enables surface textures and morphological variations unattainable with traditional formwork, shifting concrete from a structurally dominated material towards one capable of fine-grained architectural articulation. Their experiments with computationally generated panels revealed design features down to 3 mm could be faithfully reproduced, evidencing both precision and new stylistic possibilities in facade treatments. These controlled textures challenge the convention of concrete as a “blank” surface and reframe it as a medium for programmed materiality. Complementing



this, Gosselin et al. (2016) demonstrated how LFAM-enabled ultra-high-performance concrete geometries produce multifunctional wall elements where aesthetics and performance converge. By embedding thermal and acoustic functions into complex forms, the printed moulds facilitated architectural elements that operate simultaneously as expressive surfaces and engineered systems. Hybrid additive-subtractive approaches further extend these potentials: Weflen (2020) showed that CNC machining of additively manufactured moulds can refine surface quality, removing layer ridges and producing finish tolerances suitable for exposed architectural concrete, although this technique may not be possible for more complex free-form geometries.

Contemporary architectural practice increasingly relies on free-form geometries, demanding advanced mould fabrication techniques. While traditional and subtractive methods remain prevalent, their limitations in cost, waste, and sustainability are well documented. LFAM emerges as a promising alternative, addressing waste concerns through recyclability and offering significant reductions in fabrication time. Despite these advantages, challenges such as warping, structural stability, and chemical compatibility with concrete present ongoing engineering hurdles. Crucially, demoulding remains a critical bottleneck often risking damage to intricate geometries or mandating sacrificial mould use. Recent experiments, notably flexible TPU moulds (Xu et al., 2022) and circular PETG workflows (Burger et al., 2023), demonstrate effective strategies for easing the

moulding, highlighting the potential of targeted material innovations to overcome this critical obstacle. Continued research into enhanced moulding techniques will significantly expand the viability and sustainability of LFAM moulding solutions in complex architectural applications.

## Methodology & Methods.

This research adopts a design-through-research methodology grounded in action research principles, positioning iterative making as the central mode of inquiry. Rather than separating theory from practice, the project unfolds through recursive cycles of prototyping, observation, and refinement, generating knowledge through direct engagement with materials, tools, and fabrication processes. This approach is particularly suited to the demoulding challenge, which is framed not as an abstract design problem but as a situated, materially contingent constraint embedded within the act of making.

The research strategy operates within a closed-loop digital-to-physical workflow, integrating computational design, robotic fabrication, and material experimentation. Demoulding is treated as a design parameter, not a post-process concern, and is embedded into the slicing and toolpath generation stages of formwork production. A custom Grasshopper definition was iteratively developed and refined to parametrically embed seam logic into the toolpath. This script outputs KUKA Robot Language (KRL) code tailored to the specific LFAM setup at the Design Futures Lab (DFL).

Iterative prototyping was central to the methodology. Full-scale PETG moulds were printed and physically tested, with each iteration informing refinements to seam geometry, structural tolerances, print settings, and scripting logic. Speed ramping trials established the under-extrusion thresholds required to create structurally sound yet

demountable seams. Perforation strategies using start/stop commands were likewise developed and tested across multiple iterations.

All tests were documented using webcam systems mounted within the KR70 cell and on the end effector. These were connected to OBS (Open Broadcaster Software) to capture real-time print behaviour for later analysis. This footage served as a diagnostic layer, providing time-stamped visual data to assess seam formation, thermal deformation, and extrusion consistency.

Validation occurred through full-scale casting trials. PETG moulds were used in real concrete casting scenarios, with demoulding efficiency assessed via empirical observation. Removal time, tool requirements, and surface finish outcomes were documented to evaluate the seam performance.



## Case Study.

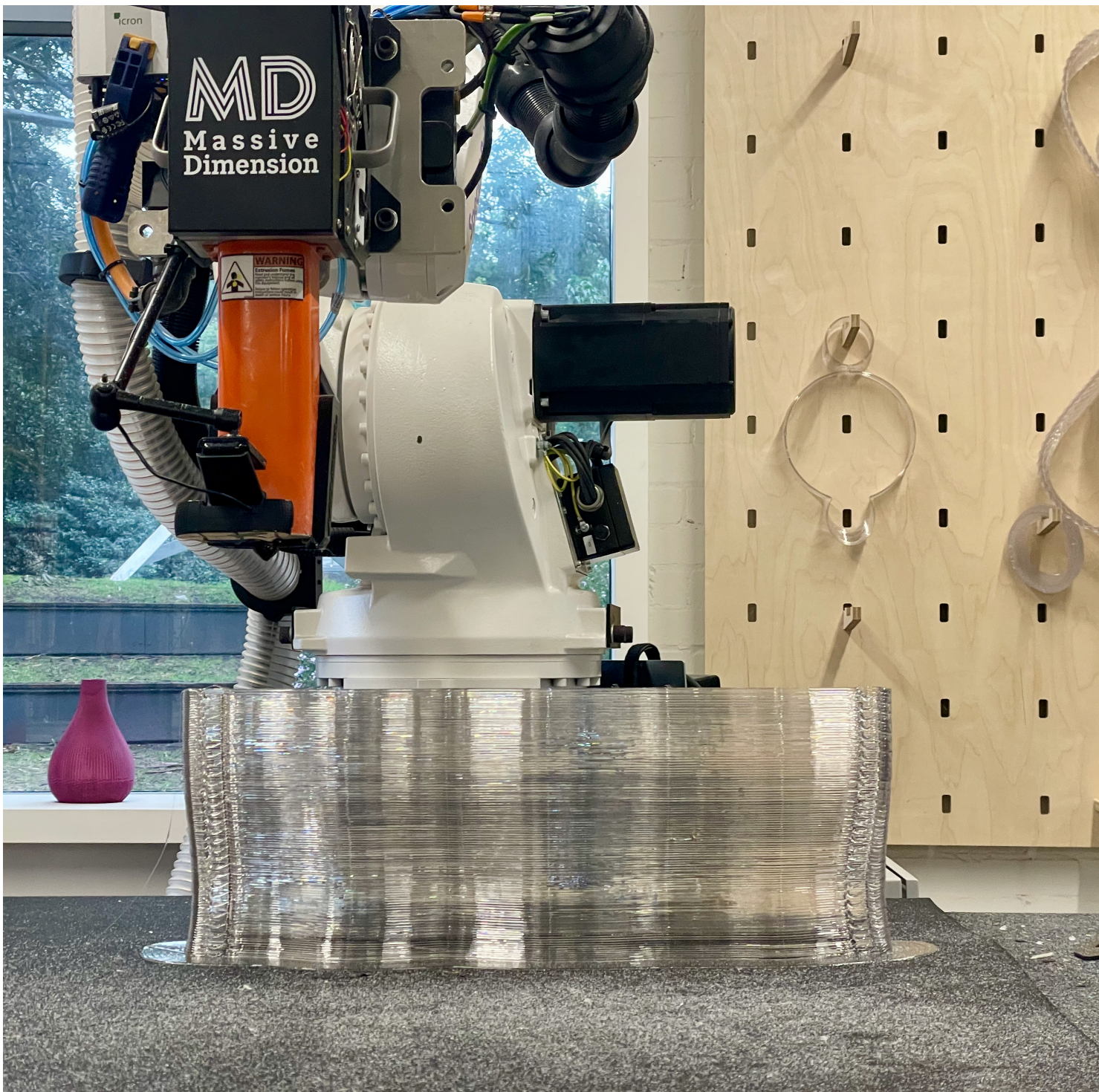


Figure 3 - KUKA KR70 and MDPE10 LFAM setup printing mould



The case study presented documents the design, development, and testing of a parametric workflow for embedding demoulding strategies into LFAM moulds. The challenging demoulding process encountered during the Bioshelter 2.0 project is the inspiration for this case study, where highly complex free-form geometries were cast in concrete via LFAM-produced PETG moulds. The cured concrete elements proved extremely difficult to demould without damaging the concrete surface. This research sought to reframe demoulding not as a destructive afterthought but as a computational design parameter. To ground the work in a relevant architectural precedent, the test geometry used in the case study was derived from the Bioshelter panel design, scaled down and simplified slightly to accelerate turnaround times and reduce concrete consumption.

The central research challenge was therefore defined: how can demoulding be integrated directly into the design-to-fabrication pipeline, rather than treated as a post-casting task? The following narrative traces the process of negotiating this question through iterative experimentation, reflecting on successes, failures, and the gradual refinement of techniques that culminated in a validated demoulding workflow.

### Negotiating Constraints

The first step in developing the workflow was to establish the technical and material constraints imposed by the DFL's fabrication environment. The LFAM system available consisted of a KUKA KR70 industrial robotic

arm fitted with an MDPE10 pellet extruder. While this setup enabled the production of large-scale PETG moulds, it was not designed for dynamic extrusion control. Extrusion occurred at a constant rate, and any attempt to alter this mid-print resulted in the robot pausing while the change was parsed. This caused unwanted plastic blobs as molten material continued to flow while the end-effector remained stationary. The only reliably modifiable parameter was robot movement speed (LIN speed), which indirectly controlled deposition size by varying how quickly the end effector moved along a given contour.

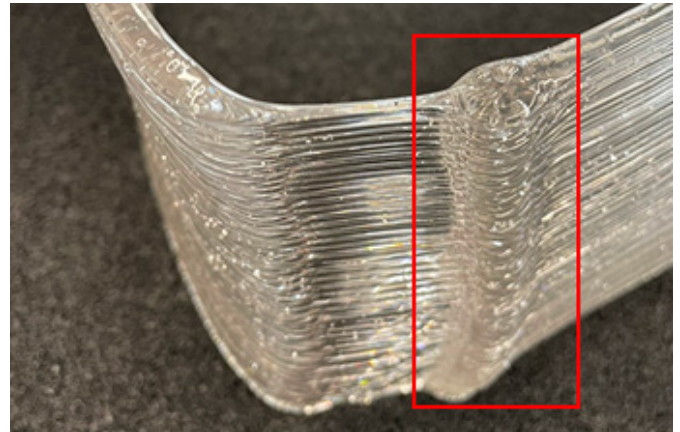


Figure 4 - "blobs" resulting from pauses in robot movement due to changing extrusion rate

These mechanical realities shaped the design strategy. Extrusion modulation could not be employed to embed weakness into the mould wall, leaving only speed control as a viable parameter for intervention. Print speed & quality was further bounded by thermal constraints. PETG requires sufficient cooling time between layers to avoid deformation, leading to the establishment of a thirty-second per layer target. This figure, calibrated through earlier DFL projects and Lamont's (2024) research ensured that each deposited layer cooled below its glass transition temperature before subsequent material was added.

Material selection was equally critical. While PLA demonstrated lower warp, it degrades rapidly in contact with the alkaline environment of wet concrete (Jipa et al., 2022). ABS offered greater durability but emits toxic fumes incompatible with the DFL's ventilation system. TPU presented interesting possibilities due to its flexibility but was unobtainable in pelletised form within the project timeframe. PETG was



Figure 5 - plastic grinder for recycling PETG

ultimately selected for its chemical resistance, strength, and compatibility with a closed-loop recycling workflow where failed prints could be shredded and re-extruded. These constraints collectively narrowed the design modulation as the primary tool for embedding demoulding behaviour.

### Early Strategies and Concept Generation

With constraints in place, several preliminary strategies were explored. Concepts such as snap-fit joinery, hinged moulds, release agent channels, and orientation-based delamination were quickly discarded. Snap-fit joints were limited to simple geometries and unsuitable for the highly curved forms of the case study. Hinges printed in PETG proved too weak to withstand hydrostatic pressure, and the material too brittle to bend properly. Attempts to exploit z-axis delamination by printing moulds on their side were impossible to print without support material, which the pellet extrusion system could not provide. Capillary channels for wax or compressed air injection



Figure 6 - interior mechanism of plastic grinder

were also impractical, given the low resolution of large format pellet extrusion. Although many of the initial strategies proved impractical, they nonetheless generated valuable insights. The hinge-based approach highlighted the fundamental challenge of balancing local fragility with global structural stability: the mould required sufficient weakness to allow for controlled destruction while maintaining adequate resistance to hydrostatic pressure of wet concrete. Similarly, explorations into release agent channels highlighted the importance of integrating demoulding capacity into the material and fabrication logic of the mould itself, rather than treating it as an additive or secondary intervention. Within this context, a “perforated tear strip” concept emerged as the most viable direction. These could be parametrically embedded across a range of geometries and provided a systematic means of introducing controlled weakness without significantly compromising the mould’s overall integrity.





Figure 7 - result of speed ramp test



Figure 8 - larger test of speed ramped seam showing that the technique works at full scales



### Speed-Ramped Under-Extrusion

The first focused line of experimentation centred on speed-ramped under-extrusion. By keeping the extrusion rate constant but accelerating the robot along a defined portion of each contour, deposition density could be strategically reduced to create weakened seams. Each layer ramped its speed at the same aligned location, ensuring a continuous seam running vertically through the print. After testing a range of multipliers, a 7x baseline speed (equating to approximately 0.2-0.35 m/s) was found to produce the most acceptable results. Seams printed at this speed were approximately 50-60% thinner than adjacent wall sections, visibly weaker and easier to cut through, yet sufficiently robust to withstand the hydrostatic pressure of curing concrete.

The method demonstrated clear successes. Seams were predictable, repeatable, and mechanically reliable. Structural stability was maintained, and the thinner sections could be cut with less effort than full-thickness walls. However, limitations soon became evident. When filled with concrete, seams became difficult to access, pressing tightly against the cast surface and leaving little space for tools. Although easier to cut than solid walls, demoulding still required chisels and saws, raising the risk of surface damage. The method marked an important step forward but did not yet provide a complete solution. The insight was valuable: speed ramping could reliably embed weakness, but accessibility remained unresolved.

### Failures and Turning Points

A parallel line of experimentation attempted to address this accessibility issue by embedding perforations along the seam. The concept was to weave extruder on/off through the KRL code either side of the seam at alternating layers, leaving small holes through which zip ties or cords could be threaded. In theory, this would allow the mould to be both reinforced and dismantled via the same system. In practice, however, molten PETG sagged into the gaps, sealing them shut and destabilising the print.

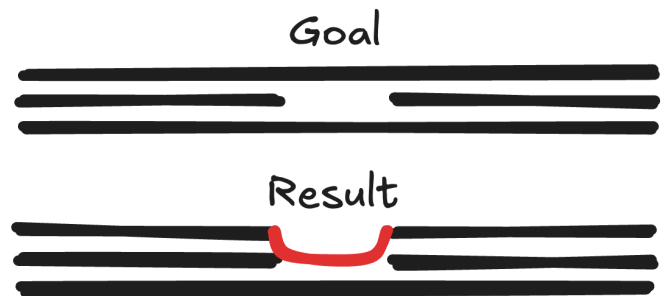


Figure 9 - diagram showing PETG sagging and destabilising prints

Yet from this failure emerged a crucial turning point. Observing the extruder's behaviour revealed two distinct phenomena. When the extruder was left on during a fast move across the seam, it produced thin, weakened strands comparable in thickness to the speed-ramped seams described previously. When it was switched off, however, the nozzle left behind hair-thin, almost invisible threads: fragile, wispy, and effectively non-structural. By alternating between extruder on/off states at high speeds, a layered sequence of weak perforations could be created. Each strand was given just enough time to slump and cool before the next pass, forming a discontinuous



Figure 10 - diagram showing PETG sagging and resulting failed print





Figure 11 - "floating" seams, both extruder on and extruder off + fast move pictured

"floating" seam that had the same ~50-60% reduction in thickness as the speed ramped trials.

This discovery reframed machine quirks as design opportunities. Rather than forcing the extruder to produce precise holes, which exceeded the system's resolution, the method exploited its natural tendency to string and slump. The result was a zipper-like seam composed of gaps distributed across layers.

### **A Parametric Workflow for Demoulding**

Building on these insights, a final workflow was developed that combined speed-ramped under-extrusion with extruder on/off perforations into a single parametric script. Implemented in Grasshopper, the script translated input geometries into structured KRL code, embedding seam placement, speed multipliers, and start/stop sequences directly into the toolpath. This workflow ensured seams were consistently generated across any geometry, transforming demoulding from a post-process task into an integral design parameter.

Though tailored to the DFL's robotic setup, the principle behind the workflow is broadly transferable. Any pellet-based LFAM system can replicate the strategy. By embedding demoulding performance into the slicing logic itself, the workflow created a replicable,

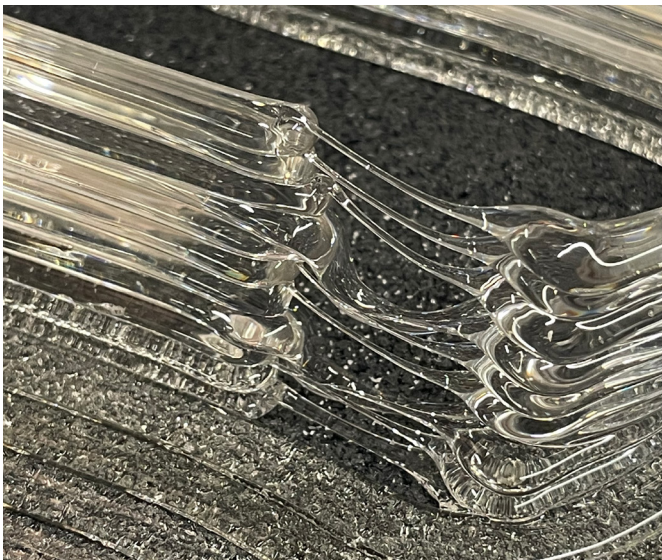


figure 12 - "floating" seams, both extruder on and extruder off + fast move pictured





Figure 13 - cast concrete structure after successful demoulding

scalable technique with potential applications far beyond the immediate case study.

### Validation Through Casting

The workflow was tested through full-scale casting trials. PETG moulds were produced with seams covered with PVC tape strips to prevent slurry ingress during casting. Once the concrete cured, seams were easily accessed with side cutters, and the mould peeled away in under thirty seconds. Unlike earlier iterations, no chisels or saws were required, and the concrete emerged completely intact with zero surface damage.

The results confirmed the repeatability of the method but also revealed new considerations. While the taped seams prevented slurry ingress, in some cases the tape adhered to the concrete surface or allowed small amounts of slurry to seep beneath, leaving minor imperfections. The tape also left a smooth surface finish along the seams, contrasted with the grooved surface of the rest of the cast concrete. These issues present the need for further research into surface finish quality when casting with LFAM-produced moulds. Nonetheless, the primary aim of rapid, non-destructive demoulding was achieved.

Figure 14 - seam covered with PVC tape to prevent concrete ingress



Figure 15 - surface finish variants due to contact with PVC tape over seams



## Discussion.

This research set out to investigate how demoulding strategies can be integrated into a parametric design-to-fabrication workflow for LFAM of mould geometries in concrete construction. The key objective was to move beyond treating demoulding as a destructive afterthought and instead embed it as a parameter within the digital and material logic of mould design. In doing so, the project aimed to address a significant bottleneck in the application of LFAM-produced moulds to complex architectural geometries.

The outcomes demonstrate that the central research aim was largely achieved. Through iterative prototyping, a parametric workflow was developed in Grasshopper that embedded demoulding capacity into robotic toolpaths by combining speed-ramped under-extrusion and extruder on/off commands. These techniques created seams that were both structurally robust under casting conditions, and sufficiently fragile to enable rapid removal. Full-scale casting trials validated the approach: PETG moulds were successfully stripped in under thirty seconds with minimal effort and no damage to the concrete element. The research therefore provides a proof-of-concept that demoulding can be parametrically encoded into LFAM workflows without compromising the geometric fidelity or stability of the moulds.

Nonetheless, several limitations emerged. While the seams enabled clean demoulding, their accessibility remained dependent on supplementary measures such as PVC tape to prevent slurry ingress. This created secondary issues: in some cases, the tape adhered to the concrete surface or produced inconsistencies in surface finish along the seam line. These

results highlight that while the structural and temporal objectives of the research were achieved, questions of surface quality and finish remain unresolved. Furthermore, the use of PETG reflects a pragmatic material choice informed by chemical resistance, availability, and recyclability, but it also constrained the exploration of alternative materials, such as TPU, that may provide intrinsic demouldability through flexibility. A further limitation lies in the technological configuration of the fabrication system. The MDPE10 pellet extruder mounted on the KR70 robotic arm lacked dynamic extrusion control, restricting experimentation to speed-based modulation. While this constraint was productively reframed into design opportunities, it prevented exploration of potentially richer extrusion strategies. Systems with real-time extrusion control could expand the range of parametric interventions, allowing for more precise and diverse demoulding features. In addition, the testing remained limited to scaled-down case studies derived from the Bioshelter 2.0 geometry. Although sufficient for proof-of-concept validation, scaling up to full architectural elements will require further investigation and testing. Despite these limitations, the research outcomes respond directly to the research question. They demonstrate that demoulding strategies can be encoded as parametric variables, shaping both toolpath logic and fabrication outcomes. In doing so, the project contributes a replicable methodology that transforms demoulding into a designable, computationally controlled property of LFAM moulds. This reframing holds significance for AEC practice, as it opens pathways for



Figure 16 - observation of the LFAM setup as it prints a final test mould



sustainable, circular workflows in which moulds can be rapidly removed, shredded, and reprinted.

Further research should extend these findings in several directions. First, systematic studies of surface quality, especially at seam interfaces, are needed to ensure that architectural finishes meet professional standards. Second, investigations into alternative thermoplastics and hybrid material systems may enhance demoulding performance. Finally, scaling the workflow to industry-level construction

will require integration with structural analysis tools to balance demouldability with the mechanical demands of large-scale concrete casting. Collectively, these directions point to a broader research agenda in which demoulding is not only resolved as a technical constraint but leveraged as a generative parameter for sustainable design-to-fabrication practice.

## Conclusion.

Integrating demoulding as a computational parameter within LFAM workflows offers a critical response to one of the most persistent limitations of 3D printed formwork for concrete construction: the difficulty of removing moulds without damage or waste. This research has addressed that challenge by developing and testing a parametric design-to-fabrication workflow that embeds demoulding logic directly into robotic toolpaths. In doing so, it reframes demoulding from a destructive, post-casting necessity into a designable feature of the mould itself.

The investigation demonstrated that demoulding strategies can be effectively encoded into toolpath generation through speed-ramped under-extrusion and extruder on/off commands, producing seams that balance structural stability with controlled weakness. Full-scale casting trials validated the efficacy of this approach, showing that PETG moulds could be removed cleanly, rapidly, and without damaging the cast object. These findings confirm that parametric demoulding workflows are not only technically viable but also scalable, offering a more sustainable and efficient alternative to sacrificial or labour-intensive mould removal practices.

While limitations remain - particularly regarding surface finish quality, material constraints, and scalability to larger architectural components, the research outcomes offer a replicable methodology that integrates computational design, robotic fabrication, and material logic into a coherent workflow, advancing sustainable approaches to architectural fabrication. The project also highlights opportunities for further exploration, including the development of alternative materials with intrinsic demouldability, hybrid additive-subtractive finishing processes, and integration with structural performance modelling.

Ultimately, this research positions demoulding not as a barrier but as a generative design parameter, expanding the potential of LFAM in both prefabricated and in-situ contexts. The key question it opens is whether demoulding can be leveraged not only as a technical solution but as a driver of new architectural expression in concrete construction.





Figure 17 - cast concrete structure with half of mould removed, showing surface finish as a result of PVC tape covering the seam

## Acknowledgements.

I would like to thank the DFL workshop staff, in particular Richard Blackwell for his expertise, mentorship, and guidance throughout this project.

I would also like to thank the Computational Design staff at UNSW, in particular Dr Nicole Gardner and Dr Cristina Ramos Jaime for their support throughout this project, as well as Ben Doherty and Daniel Yu for lighting the fire of computational design inside of me.

Finally, I would like to thank my fellow students, Jayden Tran and Louis Lamont. I could not have asked for a better group of LFAM researchers to collaborate with.



# References.

- Armstrong, K.O. et al. (2023) 'Life cycle cost, energy, and carbon emissions of molds for precast concrete: Exploring the impacts of material choices and additive manufacturing', *Resources, Conservation and Recycling*, 197, p. 107117. Available at: <https://doi.org/10.1016/j.resconrec.2023.107117>.
- Burger, J. et al. (2020) 'Eggshell: Ultra-Thin Three-Dimensional Printed Formwork for Concrete Structures', *3D Printing and Additive Manufacturing*, 7(2), pp. 48–59. Available at: <https://doi.org/10.1089/3dp.2019.0197>.
- Burger, J. et al. (2022) 'Design and fabrication of optimised ribbed concrete floor slabs using large scale 3D printed formwork', *Automation in Construction*, 144, p. 104599. Available at: <https://doi.org/10.1016/j.autcon.2022.104599>.
- Burger, J. et al. (2023) 'Circular Formwork: Recycling of 3D Printed Thermoplastic Formwork for Concrete', *Technology|Architecture + Design*, 7(2), pp. 204–215. Available at: <https://doi.org/10.1080/24751448.2023.2245724>
- Chen, H.Z., Shan, Z.D. and Dong, H.Z. (2013) 'Research of Foam Pattern Processing for Lost Foam Casting', *Applied Mechanics and Materials*, 331, pp. 600–603. Available at: <https://doi.org/10.4028/www.scientific.net/AMM.331.600>.
- Gosselin, C. et al. (2016) 'Large-scale 3D printing of ultra-high performance concrete – a new processing route for architects and builders', *Materials & Design*, 100, pp. 102–109. Available at: <https://doi.org/10.1016/j.matdes.2016.03.097>.
- Hunt, E. and G. (2023) 'An Investigation Into Water-Soluble 3D Printed Formwork Informed by Finite Element Analysis Reusable 3D Printed Formwork', in Immanuel Koh, Dagmar Reinhardt, Mohammed Makki, Mona Khakhar, Nic Bao (eds.), *HUMAN-CENTRIC - Proceedings of the 28th CAADRIA Conference*, Ahmedabad, 18-24 March 2023, pp. 251–260. CUMINCAD. Available at: [https://papers.cumincad.org/cgi-bin/works/paper/caadria2023\\_108?bypass\\_token=d2bb7b2ba6fed965502788bbcd04b010](https://papers.cumincad.org/cgi-bin/works/paper/caadria2023_108?bypass_token=d2bb7b2ba6fed965502788bbcd04b010).
- J  
ipa, A. et al. (2022) 'Environmental stress cracking of 3D-printed polymers exposed to concrete', *Additive Manufacturing*, 58, p. 103026. Available at: <https://doi.org/10.1016/j.addma.2022.103026>.
- Lamont, L. (2024) *The Need for Speed: An investigation into Parametric Velocity Control to enable variable extrusion rate on large-scale pellet- based robotic printing*. Unpublished. UNSW. Available at: [https://www.linkedin.com/posts/louis-lamont\\_the-need-for-speed-lfam-parametric-speed-activity-7346137953737330688-Z633](https://www.linkedin.com/posts/louis-lamont_the-need-for-speed-lfam-parametric-speed-activity-7346137953737330688-Z633).
- Li, W. et al. (2022) 'A review of formwork systems for modern concrete construction', *Structures*, 38, pp. 52–63. Available at: <https://doi.org/10.1016/j.istruc.2022.01.089>.
- Naboni, R. and Breseghello, L. (2020) 'High-Resolution Additive Formwork for Building-Scale Concrete Panels', in F.P. Bos et al. (eds) *Second RILEM International Conference on Concrete and Digital Fabrication*. Cham: Springer International Publishing (RILEM Bookseries), pp. 936–945. Available at: [https://doi.org/10.1007/978-3-030-49916-7\\_91](https://doi.org/10.1007/978-3-030-49916-7_91).
- Pottmann, H. (2010) 'Architectural Geometry as Design Knowledge', *Architectural Design*, 80(4), pp. 72–77. Available at: <https://doi.org/10.1002/ad.1109>.
- Pottmann, H. et al. (2015) 'Architectural geometry', *Computers & Graphics*, 47, pp. 145–164. Available at: <https://doi.org/10.1016/j.cag.2014.11.002>.
- Shawkat, S. et al. (2023) 'FREE-FORM GEOMETRY IN ARCHITECTURE', in *Proceedings of DARCH 2023 April - 4th International Conference on Architecture & Design*. DARCH 2023 April - 4th International Conference on Architecture & Design, International Organization Center of Academic Research. Available at: <https://doi.org/10.46529/darch.2023apr12>.
- Shelden, D.R. (Dennis R. (2002) *Digital surface representation and the constructibility of Gehry's architecture*. Thesis. Massachusetts Institute of Technology. Available at: <https://dspace.mit.edu/handle/1721.1/16899>.
- Weflen, E.D. (2020) *A hybrid additive and subtractive manufacturing approach for multi-material components*. Master of Science. Iowa State University. Available at: <https://doi.org/10.31274/etd-20200624-247>.
- Xu, T. et al. (2022) 'Additively manufactured thermoplastic polyurethane (TPU) mold for concrete casting of complex structure', *Rapid Prototyping Journal*, 28(9), pp. 1717–1730. Available at: <https://doi.org/10.1108/RPJ-11-2021-0307>.
- Yahya, L.M. (2023) 'Effect of Modern Technologies of Energy Conservation on Forming High-Rise Buildings', *Journal of Sustainability for Energy*, 2(3), pp. 119–131. Available at: <https://doi.org/10.56578/jse020302>.